5.5 Performing Function Operations

Name _____ Period ____ Date _____

Algebra II 5.5 (30-minute plan) · Student sheet

I can…

- add and subtract functions.
- multiply and divide functions.
- find the domain of a combined function and evaluate it.

Vocabulary · fill in the missing word

(f + g)(x) — _____

(f − g)(x) — _____

(fg)(x) — _____

(f/g)(x) — _____

Domain of a combined function — the _____ in the domain of BOTH f and g (and g(x) ≠ 0 for $\frac{f}{g}$)

Part 1 · Notes and examples (fill in each note, then work the example)

Sum and difference: (f + g)(x) = _____ (f − g)(x) = _____

Domain of a combined function: The domain of a combined function is the x-values in the domain of _____ f and g.

Example 1

Let f(x) = $3\sqrt{x}$ and g(x) = $-10\sqrt{x}$. Find (f + g)(x) and (f − g)(x) and state the domain of each.

Product and quotient: (fg)(x) = _____ (f/g)(x) = _____, where _____

Domain of f/g: For f/g, also leave out every x that makes _____.

Example 2

Let f(x) = 6x and g(x) = x^($\frac{3}{4}$). Find (fg)(x) and ($\frac{f}{g}$)(x) and state the domain of each.

Part 2 · Partner practice

Partner practice

Let $f(x) = 4x^2 - x$ and $g(x) = x^2 + 3x - 6$. Find $(f - g)(x)$ and $(f - g)(2)$.